

Notes: Costinot & Werning (RES, 2023)

Robots, Trade, and Luddism: A Sufficient Statistic Approach to Optimal Technology Regulation

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1 Big picture

- **Question.** How should governments regulate new technologies such as robots, artificial intelligence, or trade opportunities when these technologies create distributional winners and losers?
- **Core policy problem.** If lump-sum transfers or complete linear factor taxes were available, redistribution could be separated from production and taxes on robots or trade would be zero. The paper studies the realistic second-best case where income taxation is available but cannot condition on worker type.
- **Central mechanism.** Taxes on firms using the new technology can change relative wages across the wage distribution. This is *predistribution*: using production-side taxes to affect the distribution of market earnings before the income tax operates.
- **Model environment.** The economy has heterogeneous households, many goods, old and new technologies, nonlinear income taxation, and ad valorem taxes on new-technology firms.
- **Main theoretical contribution.** The paper derives optimal technology-tax formulas using sufficient statistics rather than a fully structural model of tasks, trade, robots, or production networks.
- **Key sufficient statistic.** The most important statistic is how a marginal change in new-technology output changes the *growth rate of wages across the wage distribution*.
- **Formula 1.** Proposition 1 studies tax changes holding the income tax schedule fixed. It contains social welfare weights and generalizes political-economy formulas for tariffs.
- **Formula 2.** Proposition 2 studies tax changes accompanied by income-tax adjustments that keep the utility distribution fixed. It eliminates welfare weights and leads to an empirical sufficient-statistic formula.
- **Empirical implementation.** Using evidence on robots and Chinese import exposure, the efficient tax on robots is several percentage points, while the efficient tariff on Chinese imports is tiny.
- **Comparative statics.** Even if robots or trade increase inequality, the optimal tax on them may fall as automation or globalization deepens.
- **Takeaway.** Distributional concerns justify nonzero taxes on robots and trade in second best, but more inequality does not mechanically imply more Luddism or protectionism.

2 Model primitives and measurement

2.1 Households and preferences

There is a continuum of heterogeneous households indexed by skill vector $\theta \in \Theta \subseteq \mathbb{R}^K$, distributed according to F . Households share weakly separable preferences over goods and labour:

$$U(\theta) = u(C(\theta), n(\theta)), \quad C(\theta) = v(c(\theta)), \quad (1)$$

where $c(\theta) = \{c_i(\theta)\}$ is the vector of goods consumed, $C(\theta)$ is subutility from consumption, and $n(\theta)$ is labour supply.

- The upper-level utility $u(C, n)$ is quasi-concave, increasing in C , and decreasing in n .
- The lower-level utility $v(c)$ is quasi-homothetic, quasi-concave, and increasing in goods.
- Skill heterogeneity can be multidimensional; the paper does not require a one-dimensional Mirrlees type.

Interpretation: The model is general enough to include Roy-style comparative advantage and task assignments. Workers can be more productive in some activities and less productive in others.

2.2 Old technology

Old technology firms produce a vector of net outputs $y = \{y_i\}$ using labour demand schedule $n = \{n(\theta)\}$. Their production set is described by

$$G(y, n) \leq 0, \quad (2)$$

where G is increasing, convex, and homogeneous in (y, n) .

- Positive y_i denotes output and negative y_i denotes an input.
- Homogeneity gives constant returns to scale.
- The old technology can include arbitrary production networks and global supply chains.

Interpretation: The old technology is kept abstract so that the final tax formulas do not depend on whether robots replace tasks, trade changes sourcing patterns, or production networks reallocate inputs.

2.3 New technology

New technology firms produce net output $y^* = \{y_i^*\}$ satisfying

$$G^*(y^*; \phi) \leq 0, \quad (3)$$

where $\phi > 0$ is a productivity or opportunity parameter.

- In the robot application, new technology firms can be robot producers that transform goods into machines.
- In the trade application, new technology firms can be traders who export and import goods.
- In a trade interpretation, one can write

$$G^*(y^*; \phi) = \sum_i \bar{p}_i(\phi) y_i^*, \quad (4)$$

where $\bar{p}_i(\phi)$ are world prices.

Interpretation: The new technology does not directly employ labour. It affects wages indirectly by changing the production environment faced by old-technology firms.

2.4 Resource constraint

For all goods,

$$\int c(\theta) dF(\theta) = y + y^*. \quad (5)$$

Interpretation: Aggregate household demand equals the combined supply of old and new technology firms.

2.5 Prices, income taxes, and technology taxes

Let $w(\theta)$ be the wage paid to skill type θ . Because the government uses nonlinear income taxation, a household with earnings $w(\theta)n(\theta)$ retains

$$w(\theta)n(\theta) - T(w(\theta)n(\theta)). \quad (6)$$

Let p_i be the price faced by old-technology firms and households, and let p_i^* be the price faced by new-technology firms. Ad valorem taxes on new technology firms are t_i^* , so non-arbitrage requires

$$p_i = (1 + t_i^*)p_i^*, \quad \text{for all } i. \quad (7)$$

The normalization is

$$p_1 = p_1^* = 1, \quad t_1^* = 0. \quad (8)$$

Interpretation: Technology regulation is captured by wedges between old-technology and new-technology prices. A robot tax, import tariff, export subsidy, or import subsidy is represented by a component of t^* .

3 Equilibrium, welfare, and government problem

3.1 Competitive equilibrium

An equilibrium consists of allocations, prices, wages, income taxes, and technology taxes such that:

1. households maximize utility;
2. old and new technology firms maximize profits;
3. goods and labour markets clear;
4. non-arbitrage conditions hold;
5. the government budget is balanced.

Profit maximization by old technology firms implies a wage schedule

$$w = w(p, n; \theta). \quad (9)$$

Interpretation: The dependence of wages on goods prices and labour allocation is the bridge between technology taxes and redistribution.

3.2 Social welfare

Let $\bar{U}(z)$ denote the utility of the household at utility quantile $z \in [0, 1]$. Social welfare is

$$W(\bar{U}), \quad (10)$$

where W is strictly increasing in the distribution of utility.

Interpretation: The welfare function depends on the distribution of well-being, not directly on identities. This allows the paper to collapse multidimensional heterogeneity into quantiles of utility or wages.

3.3 Government problem

The government selects a competitive equilibrium to maximize social welfare:

$$\max_{(c, n, y, y^*, p, p^*, w, T, t^*, \bar{U}) \in \Omega} W(\bar{U}) \quad (11)$$

subject to

$$G^*(y^*; \phi) = 0. \quad (12)$$

By Walras' Law, this condition is equivalent to government budget balance.

Interpretation: The government chooses both the income tax schedule and new-technology taxes. The question is what technology taxes complement optimal income taxation.

4 Optimal technology regulation

4.1 Efficiency versus redistribution: Lemma 1

The paper considers marginal changes δt^* in technology taxes and δT in the income tax schedule. Let $\bar{w}(z)$, $\bar{n}(z)$, $\bar{x}(z)$, and $\bar{c}(z)$ denote wage, labour supply, earnings, and consumption at quantile z . Let $\tau(z) = T'(\bar{x}(z))$.

Lemma 1 states that at the optimum, for any feasible variation,

$$\begin{aligned} & - \sum_i t_i^* (p_i^* y_i^*) \delta \ln y_i^* - \int \tau(z) \bar{x}(z) \delta \ln \bar{n}(z) dz \\ & = \int [\bar{\lambda}(z) - 1] \bar{x}(z) \left[(1 - \tau(z)) \delta \ln \bar{w}(z) - \frac{\delta T(z)}{\bar{x}(z)} - \sum_i \frac{p_i \bar{c}_i(z)}{\bar{x}(z)} \delta \ln p_i \right] dz. \end{aligned} \quad (13)$$

- The left side is the efficiency cost: the technology-tax fiscal externality plus the income-tax fiscal externality.
- The right side is the redistribution benefit: changes in real income weighted by social marginal welfare weights.

Interpretation: Lemma 1 is the master equation. All optimal technology-tax formulas are different ways of choosing variations in t^* and T .

4.2 Proposition 1: fixed income tax schedule

The first variation holds the shape of the income tax schedule fixed, $\delta T = 0$, up to a lump-sum transfer. The resulting formula is

$$\begin{aligned} t_i^* &= \int \left(1 - \frac{\bar{\lambda}(z)}{\int \bar{\lambda}(v) dv} \right) \frac{\bar{x}(z)}{p_i^* y_i^*} \left[(1 - \tau(z)) \frac{\delta \ln \bar{w}(z)}{\delta \ln y_i^*} - \sum_j \frac{p_j \bar{c}_j(z)}{\bar{x}(z)} \frac{\delta \ln p_j}{\delta \ln y_i^*} \right] dz \\ & - \int \tau(z) \frac{\bar{x}(z)}{p_i^* y_i^*} \frac{\delta \ln \bar{n}(z)}{\delta \ln \bar{w}(z)} \frac{\delta \ln \bar{w}(z)}{\delta \ln y_i^*} dz. \end{aligned} \quad (14)$$

Interpretation: This formula contains welfare weights. It asks whether the technology tax moves real income toward households with high social marginal value, after accounting for fiscal externalities.

4.3 Corollary: political-economy tariffs as predistribution

With quasi-linear preferences, inelastic labour supply, no labour income taxation, and sector-specific factors, the formula simplifies to

$$t_i^* = \left(\frac{\bar{\lambda}(i)}{\int \bar{\lambda}(v) dv} - 1 \right) \left(- \frac{1}{p_i^*} \frac{d\bar{x}(i)}{dy_i^*} \right). \quad (15)$$

Interpretation: Protection in a sector equals the weight placed on that sector's workers times the marginal effect of lower imports on their earnings. Many political-economy tariff formulas are special cases with different welfare weights.

4.4 Proposition 2: utility-preserving income tax adjustment

The second variation adjusts the income tax schedule so that the distribution of utility remains fixed, $\delta \bar{U} = 0$. The optimal tax formula is

$$t_i^* = \int \tau(z) \frac{\bar{x}(z)}{p_i^* y_i^*} \frac{\varepsilon_H(z)}{\varepsilon_M(z) + 1} \frac{\delta \ln \omega(z)}{\delta \ln y_i^*} \bigg|_{\delta \bar{U}=0} dz, \quad (16)$$

where

$$\omega(z) = \frac{\bar{w}'(z)}{\bar{w}(z)} \quad (17)$$

is the growth rate of wages across quantiles.

Interpretation: This formula is attractive because it does not require welfare weights. The relevant objects are marginal tax rates, earnings shares, labour supply elasticities, and the effect of the technology on wage inequality.

4.5 Why wage growth matters

The key incentive constraint is

$$\bar{U}(z) = \max_{z'} u \left(\bar{C}(z'), \bar{n}(z') \frac{\bar{w}(z')}{\bar{w}(z)} \right). \quad (18)$$

The envelope theorem implies that holding the utility distribution fixed requires

$$\delta \ln \bar{n}(z) = - \frac{\varepsilon_H(z)}{\varepsilon_M(z) + 1} \delta \ln \omega(z). \quad (19)$$

Interpretation: If a technology changes the slope of the wage schedule, incentive-compatible labour supply must adjust. The resulting income-tax revenue effect determines the optimal technology tax.

4.6 Proposition 3: empirical approximation

Because the utility-preserving wage response may not be directly observed, the paper shows that a budget-balanced wage response can be used to first order:

$$t_i^* = \int \tau(z) \frac{\bar{x}(z)}{p_i^* y_i^*} \frac{\varepsilon_H(z)}{\varepsilon_M(z) + 1} \frac{\delta \ln \omega(z)}{\delta \ln y_i^*} \Big|_{\delta G^*=0} dz + O(\bar{\varepsilon}^2). \quad (20)$$

Interpretation: Small labour supply elasticities make the approximation empirically useful. Researchers can use reduced-form estimates of how robots or import exposure shift the wage distribution.

5 Putting the formulas to work

5.1 Efficient tax on robots

The implementation uses Proposition 3 and evidence from Acemoglu and Restrepo on how exposure to robots affects wages across the wage distribution. The relevant input is

$$\frac{\delta \ln \omega(z)}{\delta \ln y_m^*}, \quad (21)$$

where y_m^* is the quantity of robots per thousand workers.

- The wage semi-elasticities are estimated across wage deciles.
- Marginal income tax rates, labour supply elasticities, and the share of robots in the economy complete the formula.
- The baseline efficient robot tax is 4.28%.
- Alternative elasticities and assumptions produce values around 3.83% to 6.42%.

Interpretation: Robots are a small but nontrivial sector. If robots disproportionately raise high-wage earnings, the fiscal externality through income taxation justifies a positive robot tax.

5.2 Efficient tariff on Chinese imports

The implementation for trade uses Chetverikov, Larsen, and Palmer's estimates of how Chinese import exposure affects the wage distribution. The same sufficient statistic formula applies to import exposure.

- Wage effects from Chinese imports are similar in magnitude to robot wage effects.
- But Chinese imports are a much larger share of the economy.
- Because the formula scales by $p_i^* y_i^*$, a similar wage effect over a larger base implies a smaller tax rate.
- The baseline efficient tariff is only 0.07%.

Interpretation: The comparison between robots and trade shows why policy magnitudes depend not only on wage effects but also on the scale of the activity being taxed.

6 Comparative statics

6.1 A simple economy with machines

The paper then studies a closed-form economy with one final good and one intermediate good, machines. Households have one-dimensional skill $\theta \in [0, 1]$ and quasi-linear preferences:

$$U(\theta) = C(\theta) - \frac{n(\theta)^{1+1/\varepsilon}}{1+1/\varepsilon}. \quad (22)$$

Old technology produces final output using workers and machines:

$$G(y_f, y_m, n) = y_f - \max_{\{\tilde{y}_m(\theta)\}} \left\{ \int g(\tilde{y}_m(\theta), n(\theta); \theta) dF(\theta) \mid \int \tilde{y}_m(\theta) dF(\theta) \leq -y_m \right\} \leq 0, \quad (23)$$

with

$$g(\tilde{y}_m, n; \theta) = \exp(\alpha(\theta)) \left(\frac{\tilde{y}_m(\theta)}{\beta(\theta)} \right)^{\beta(\theta)} \left(\frac{n(\theta)}{1 - \beta(\theta)} \right)^{1 - \beta(\theta)}. \quad (24)$$

New technology produces machines:

$$G^*(y_f^*, y_m^*; \phi) = \phi y_f^* + y_m^*. \quad (25)$$

Interpretation: More skilled workers use machines more intensively, so cheaper machines raise inequality.

6.2 Equilibrium wage schedule

Profit maximization implies

$$p_m^* = \frac{1}{\phi}, \quad (26)$$

and

$$w(p_m; \theta) = (1 - \theta)^{-1/\gamma(p_m)}, \quad \gamma(p_m) = \frac{1}{\alpha - \beta \ln p_m}. \quad (27)$$

The effect of machine prices on wage inequality is

$$\frac{d \ln \omega(\theta)}{d \ln p_m} = -\frac{d \ln \gamma(p_m)}{d \ln p_m} = -\beta \gamma(p_m) < 0. \quad (28)$$

Interpretation: If machines become cheaper, skilled workers who use machines more intensively benefit more, so the wage distribution becomes more unequal.

6.3 Social welfare in the simple economy

The government maximizes

$$W(U) = \int U(\theta) d\Gamma(\theta), \quad (29)$$

with

$$\Gamma(\theta) = \lambda + \theta(1 - \lambda). \quad (30)$$

The parameter $\lambda \in [0, 1]$ controls redistributive preferences:

- $\lambda = 0$ gives utilitarian weights;
- $\lambda = 1$ gives a Rawlsian limit that focuses on the poorest type.

6.4 Optimal machine tax

In the simple economy, the optimal tax satisfies

$$\frac{t_m^*}{1 + t_m^*} = \frac{\frac{\varepsilon}{\varepsilon+1} \frac{d \ln \omega}{d \ln y_m^*} \tau^* (1 - s_m)}{1 - \frac{\varepsilon}{\varepsilon+1} \frac{d \ln \omega}{d \ln y_m^*} \tau^* s_m}, \quad (31)$$

where s_m is the share of machines in gross output.

Interpretation: The numerator captures the fiscal externality created by machine-induced wage inequality. The denominator accounts for the tax base and general equilibrium feedback.

6.5 Proposition 4: technological progress lowers the optimal tax

In the simple economy, the optimal tax on machines is decreasing in the productivity ϕ of new technology firms.

Interpretation: Even though cheaper machines increase inequality, the marginal fiscal externality per machine can fall. Thus deeper automation or globalization may optimally be met with lower taxes on machines or trade.

6.6 Proposition 5: social preferences raise technology taxes

When the government cares more about the poor, the optimal and constrained optimal taxes on machines rise with λ .

When income taxes are exogenously set at a linear schedule T^c , the constrained optimal tax is

$$\frac{t_m^c}{1 + t_m^c} = \frac{[\lambda(1 - \tau^c) - \tau^c \varepsilon][1 - (1 + \varepsilon)/\gamma(p_m)]}{2\beta(1 + \varepsilon) + 1 - (1 + \varepsilon)/\gamma(p_m)}. \quad (32)$$

Interpretation: If the income tax schedule is too progressive, the constrained optimal technology policy may even subsidize machines to undo excessive redistribution through income taxation.

7 Identification and empirical use of the theory

7.1 What the sufficient-statistic approach requires

To implement Proposition 3, one needs:

1. the marginal income tax rate $\tau(z)$ across wage quantiles;
2. earnings $\bar{x}(z)$ across quantiles;
3. Hicksian and Marshallian labour supply elasticities;
4. the size of the technology sector, $p_i^* y_i^*$;
5. the estimated effect of the technology on wage growth across quantiles.

Interpretation: The approach deliberately avoids specifying the full production structure. The empirical burden is shifted to measuring wage-distribution effects.

7.2 Why reduced-form wage estimates are enough

The formula does not require knowing whether robots affect wages through task replacement, capital deepening, or changes in product demand. It only requires the marginal wage response.

Interpretation: This is the central advantage over structural quantitative models: the formula converts empirical wage-distribution effects directly into policy recommendations.

7.3 What remains imperfect

- Reduced-form estimates may be local and may not represent the marginal policy variation exactly.
- Labour supply elasticities are uncertain.
- General equilibrium responses across goods and sectors are summarized only through wage effects.
- The approach focuses on distributional/fiscal externalities, not political economy, innovation incentives, or international retaliation.

Interpretation: The formulas are powerful but not automatic. The quality of the policy conclusion depends on the quality of wage-effect and elasticity estimates.

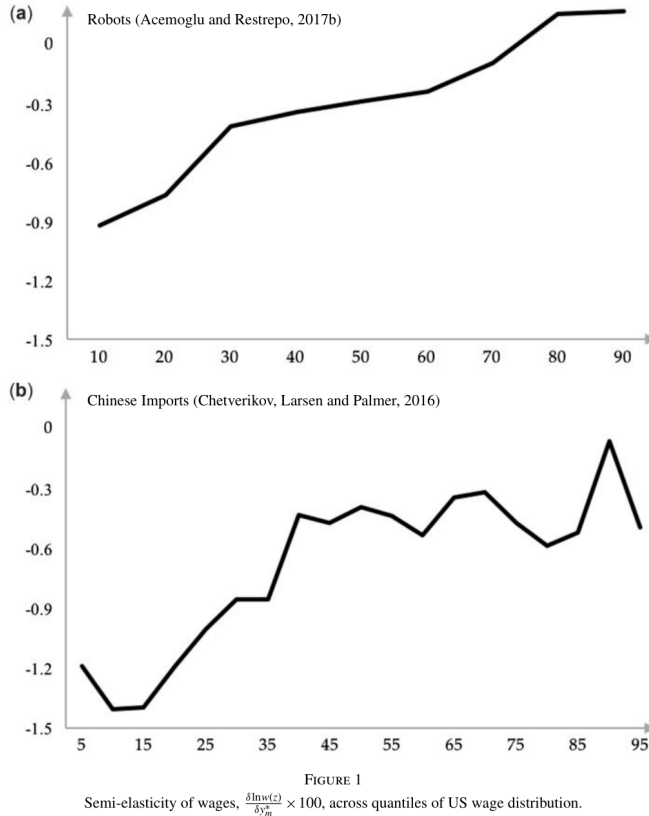
8 Tables and figures

8.1 Figure 1 and Table 1: empirical wage effects and efficient taxes

Read:

- Figure 1(a) reports the estimated semi-elasticity of wages with respect to robots across wage deciles, based on Acemoglu and Restrepo.
- Figure 1(b) reports the analogous wage semi-elasticity for Chinese imports, based on Chetverikov, Larsen, and Palmer.
- Table 1 uses these wage effects and Proposition 3 to compute efficient taxes on robots and Chinese imports.
- The baseline robot tax is 4.28%, with alternative calculations ranging from 3.83% to 6.42%.
- The baseline Chinese import tax is 0.07%, with alternative calculations ranging from 0.02% to 0.19%.

Economic message: Similar wage-distribution effects can imply very different tax rates because the size of the taxed activity differs. Robots are a small sector, so a given wage externality implies a larger ad valorem tax than for broad import flows.



(a) Figure 1: wage semi-elasticities for robots and Chinese imports.

	Baseline	$\tau_M = 0$	Effective τ	Fitted $\frac{\delta \ln w}{\delta y_M}$	Fitted $\frac{\delta \ln w}{\delta y_M}$
	(1)	(2)	(3)	(4)	(5)
Robots tax:					
τ_M^*	4.28%	6.42%	3.83%	5.38%	4.49%
Chinese imports tax:					
τ_M^*	0.07%	0.10%	0.12%	0.02%	0.19%

Notes: Column (1) reports the optimal taxes on robots and Chinese imports obtained from equation (8) under our baseline calibration. Column (2)–(5) report the same taxes using instead high values of the labour supply elasticities ϵ_M and ϵ_M , effective marginal tax rates $\tau(z)$, fitted values of the semi-elasticities $\frac{\delta \ln w}{\delta y_M}$ and elasticities $\frac{\delta \ln w}{\delta y_M}$, respectively, as further

(b) Table 1: efficient taxes.

8.2 Figure 2: social preferences, income taxes, and technology regulation

Read:

- Panel (a) compares the fully optimal machine tax t_m^* with the constrained optimal machine tax t_m^c as social preferences become more redistributive.

- The fully optimal tax is small and relatively stable, moving from 0% in the utilitarian case to about 5.43% in the Rawlsian case.
- The constrained optimal tax is much more sensitive because machine taxes must compensate for a fixed income tax schedule.
- Panel (b) reports welfare gains from optimal and constrained optimal taxation.

Economic message: Technology taxes are modest when income taxes are optimally adjusted. If the income tax schedule is constrained, however, technology taxes may be used more aggressively as a substitute redistributive instrument.

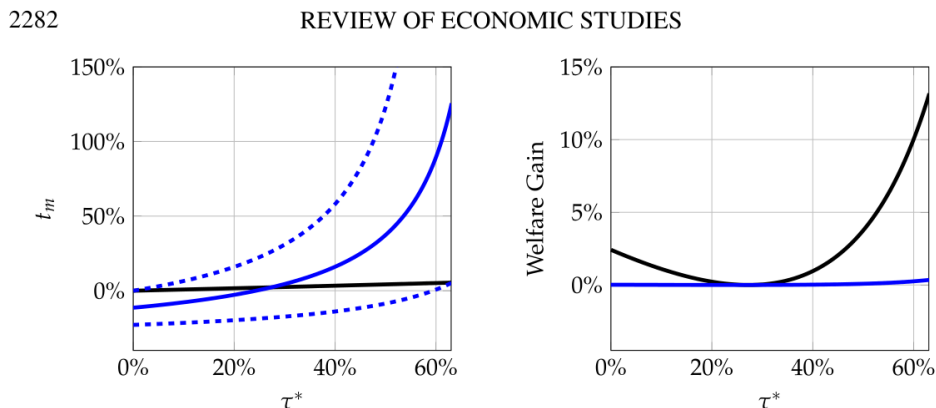


FIGURE 2
Social preferences, income taxes, and technology regulation

9 Short summary

- The paper studies optimal regulation of new technologies such as robots and trade in a second-best tax system.
- With complete redistribution tools, production efficiency would imply zero robot taxes and free trade.
- With nonlinear income taxes that cannot condition on worker type, technology taxes can improve redistribution by changing wages before taxes.
- Lemma 1 decomposes any technology-tax variation into efficiency costs and redistribution benefits.
- Proposition 1 gives a formula with welfare weights when the income tax schedule is held fixed.
- Proposition 2 gives a welfare-weight-free formula when the income tax schedule is adjusted to keep utility quantiles fixed.
- Proposition 3 shows how to use budget-balanced wage responses to approximate the formula empirically.
- Using robot evidence, the efficient robot tax is several percentage points.
- Using Chinese import evidence, the efficient import tariff is tiny.
- In the comparative-static model, the optimal technology tax can fall as automation or globalization deepens, even if inequality rises.
- The paper's main message is that distributional concerns justify some technology regulation, but not necessarily more regulation as technology advances.

10 Conclusion

10.1 Core conclusion

Costinot and Werning provide a general sufficient-statistic theory of optimal technology regulation. In a second-best world with nonlinear income taxation but no worker-type-specific transfers, taxes on robots or trade may be desirable because they affect the wage distribution and therefore the fiscal externalities created by income taxation.

10.2 Economic intuition

New technologies change relative wages. If a technology raises high-wage earnings relative to low-wage earnings, income taxes raise less revenue or require more distortion to achieve redistribution. A tax on the technology can reduce this fiscal externality. The optimal tax depends not on the details of production, but on how the technology changes wage growth across quantiles, how much income taxes distort labour supply, and the scale of the taxed sector.

10.3 Takeaway

The paper reframes robot taxes and protectionism as second-best predistribution instruments. It also cautions against a simple anti-technology conclusion: as automation and globalization expand, optimal taxes on those technologies may decline. The right policy response depends on sufficient statistics, not slogans about robots, trade, or Luddism.